Postpartum Complications

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Learning Objectives

- Identify the causes, signs and symptoms, and medical and nursing management of postpartum hemorrhage
- Describe hemorrhagic shock as a complication of postpartum hemorrhage including management and hazards of therapy
- Differentiate the causes of postpartum infection
- Summarize assessment and care of women with postpartum infection
- Describe thromboembolic disorders, including incidence, etiology, signs and symptoms, and medical and nursing management

Postpartum Hemorrhage (PPH)

- Definition and incidence
  - Traditionally defined as follows:
    - Loss of 500 ml of blood after vaginal birth
    - Loss of 1,000 ml after cesarean birth
    - A 10% change in Hct between labor and postpartum
  - Leading cause of maternal morbidity and mortality
  - Often unrecognized until mother has profound symptoms
  - PPH is classified as early or late with respect to the birth.
    - Early, acute, or primary PPH occurs within 24 hours of the birth
    - Late or secondary PPH occurs more than 24 hours but less than 6 weeks after the birth.

Remember that during pregnancy, mom's blood volume doubles to 1) provide adequate blood flow for the baby and 2) because physiologically they will lose blood after delivery so their body compensates and makes extra.

Postpartum Hemorrhage Causes

UTERINE ATONY

- Retained placenta
  - Non-adherent retained placenta
  - Adherent retained placenta
- Lacerations of genital tract
- Hematomas

OTHER CAUSES

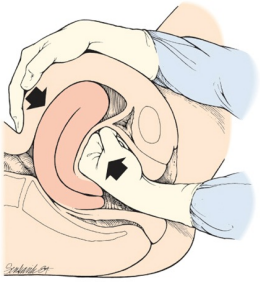
- Marked hypotonia of uterus
- Leading cause of PPH
- Associated with
  - High parity
  - Hydramnios
  - Macrosomic fetus
  - Multifetal gestation

Basically anything that will overdistend the uterus, could cause uterine atony.
Also complicated by their short postpartum stays—MUST educate on normal amount of bleeding vs. s/s of a hemorrhage.
Dark red blood=venous, superficial;
Bright red blood=arterial, deeper
Cervix and uterus are very vascular and will have bright red bleeding if there is a cervical laceration and can cause PPH
TX: suture, evacuation, ice, pericare

Postpartum Hemorrhage Causes

OTHER CAUSES

- Inversion of the uterus
  - Potentially life threatening
  - Occurs in 1 in 2500 births
- Subinvolution of the uterus
  - Late postpartum bleeding
- Placenta Accreta



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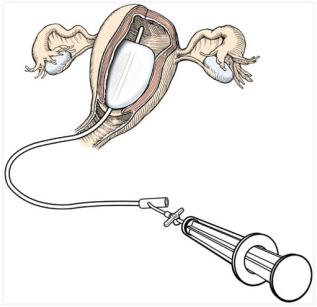

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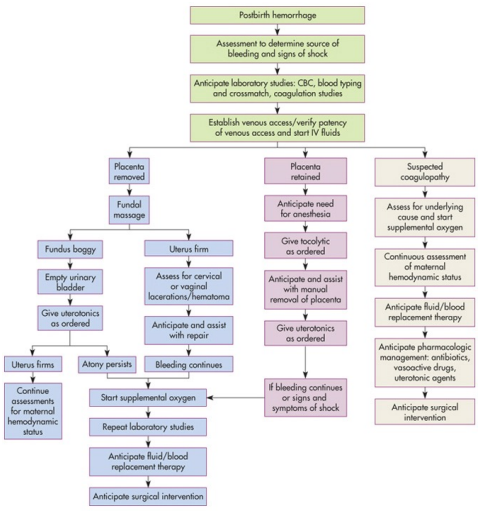
Uterine inversion is a potentially fatal complication of childbirth. Uterine inversion means the placenta fails to detach from the uterine wall, and pulls the uterus inside-out as it exits. Could be caused by excessive pressure on a relaxed uterus (uterine massage) or excessive traction on the cord during delivery.

Subinvolution means the uterus isn't decreasing in size as it should

Placenta accreta—when the placenta grows too deeply into the uterine wall

Left pic: Bimanual massage; provide sedation if no epidural
Right pic: Bakri balloon—filled with saline to apply pressure and tamponade the uterine vessels to stop the bleeding; bed rest and NO fundal massage; can be connected to a foley bag to measure the blood loss

Postpartum Hemorrhage Care Management



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Postpartum Hemorrhage Care Management

- Early recognition and treatment of PPH are critical.
- The initial intervention is firm massage of the fundus.
- Expression of any clots in the uterus
- Elimination of bladder distention
- Continuous intravenous (IV) infusion of 10 to 40 units of oxytocin added to 1,000 ml IV fluid
- Additional uterotonic medications

Methergine or Hemabate
Cytotec given rectally

Hemorrhagic (Hypovolemic) Shock

- Results from hemorrhage
  - Emergency situation in which perfusion of organs may become severely compromised
  - Can lead to death if not treated quickly
- Medical management
  - Restore circulating blood volume
- Nursing interventions
  - Monitor pulse and blood pressure
- Fluid or blood replacement therapy

Find the source/cause of the bleeding
Women with PPH can go into DIC quickly so we need to draw a DIC panel
X 2 Ivs (one for IV fluids, one for blood)
Want a minimum of 30 ml of urine out or we suspect kidney decompensation

Coagulopathies

- Idiopathic thrombocytopenic purpura (ITP)
  - Autoimmune disorder in which antiplatelet antibodies decrease the life span of platelets
- von Willebrand disease (vWD)
  - A type of hemophilia
  - Deficiency or defect in blood clotting protein

ITP: low platelets, higher risk of bleeding; tx with corticosteroids closer to delivery or a platelet transfusion

vWD: one of the more common congenital bleeding disorders; about 50% of women have a PPH so we plan for it if this disease is known; tx is desmopressin—similar to vasopressin, given right before delivery to decrease the chances of PPH

Venous Thromboembolic Disorders

- Venous thromboembolism (VTE): results from formation of blood clot or clots inside a blood vessel, caused by inflammation or partial obstruction of the vessel
  - Superficial venous thrombosis: involvement of the superficial saphenous venous system
  - Deep venous thrombosis (DVT): occurs most often in the lower extremities; involvement varies but can extend from the foot to the iliofemoral region
  - Pulmonary embolism (PE): complication of DVT occurring when part of a blood clot dislodges and is carried to the pulmonary artery, where it occludes the vessel and obstructs blood flow to the lungs

VTEs are very common in pregnancy and the postpartum period

Venous Thromboembolic Disorders

- Incidence and etiology
  - 1 per 1,000 pregnancies
  - C-section moms are at 2x risk
  - Major causes: venous stasis and hypercoagulation
- Clinical manifestations
  - Superficial venous thrombosis (most common form) is characterized by pain and tenderness in the lower extremity
- Medical management
- Nursing interventions

Most VTEs occur in the first 3 weeks after birth; DVTs are common during pregnancy, but PEs happen after birth

Tx: anticoagulants (prefer Lovenox as it is safer when breastfeeding), no razors, soft toothbrush, check pulses

Birth control pills?

Postpartum Infections

- Also called *puerperal infection*
- Any clinical infection of the genital tract that occurs within 28 days after miscarriage, induced abortion, or birth
- Defined as presence of a fever of 100.4° F (38° C) or more on 2 successive days of the first 10 postpartum days
  - Do not count the first 24 hours after birth

Most common in women with other complications/comorbidities like DM, HTN…etc.
Risk factors for infection?

Postpartum Infections

- Endometritis
  - Infection of the lining of the uterus
  - Most common postpartum infection
- Wound infections
  - Often develop after mothers are discharged home
  - Typically cesarean incision, repaired laceration, or episiotomy site
- Urinary tract infections

Endometritis—flu like symptoms, change in lochia, foul smelling discharge
REEDA
Prevention of infections is KEY!

Post-Birth Warning Signs

SAVE YOUR LIFE:

Get Care for These POST-BIRTH Warning Signs

Most women and postpartum people who give birth recover without problems. **But anyone can have a complication for up to one year after birth.** Learning to recognize these POST-BIRTH warning signs and knowing what to do can save your life.

Trust your instincts. It's always per medical care if you are not feeling well or have questions or concerns.

Call 911 if you have:	❑ **P**ain in chest ❑ **O**bstructed breathing or shortness of breath ❑ **S**eizures ❑ **T**houghts of hurting yourself or someone else
Call your healthcare provider if you have: (you only need one sign) (If you can't reach your healthcare provider, call 911 or go to an emergency room)	❑ **B**leeding, soaking through one pad/hour, or blood clots, the size of an egg or bigger ❑ **I**ncision that is not healing ❑ **R**ed or swollen leg, that is painful or warm to touch ❑ **T**emperature of 100.4°F or higher or 96.8°F or lower ❑ **H**eadache that does not get better, even after taking medicine, or bad headache with vision changes

Tell 911 or your healthcare provider: "I gave birth on _____ (Date) and I am having _____ (Specific warning signs)."

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These post-birth warning signs can become life-threatening if you don't receive medical care right away because:

- **Pain in chest, obstructed breathing or shortness of breath** (trouble catching your breath) may mean you have a blood clot in your lung or a heart problem
- **Seizures** may mean you have a condition called eclampsia
- **Thoughts or feelings of wanting to hurt yourself or someone else** may mean you have postpartum depression
- **Bleeding (heavy)**, soaking more than one pad in an hour or passing an egg-sized clot or bigger may mean you have an obstetric hemorrhage
- **Incision that is not healing, increased redness or any pus** from episiotomy, vaginal tear, or C-section site may mean an infection
- **Redness, swelling, warmth, or pain** in the calf area of your leg may mean you have a blood clot
- **Temperature of 100.4°F or higher or 96.8°F or lower**, bad smelling vaginal blood or discharge may mean you have an infection.
- **Headache (very painful), vision changes, or pain in the upper right area of your belly** may mean you have high blood pressure or post birth preeclampsia

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Key Points

- Postpartum hemorrhage is a major cause of obstetric morbidity and mortality throughout the world and is the leading reason for obstetric ICU admissions
- #1 cause of postpartum hemorrhage is uterine atony
- Hemorrhagic (hypovolemic) shock is an emergency situation in which the perfusion of body organs can become severely compromised and death can ensue
- Clotting disorders are associated with many obstetric complications
- Postpartum infection is a major cause of maternal morbidity and mortality throughout the world
- Postpartum UTIs are common because of trauma experienced during labor
- Prevention is the most effective and least expensive treatment of postpartum infection